

# Comparing More Than 2 Groups

How do we do statistical testing with more than 2 groups?

**ANOVA!**

## One-Way ANOVA

### **Null Hypothesis**

The means of all groups are the same.

**Alternative Hypothesis** At least one group mean is different.

## 2 – Sample T-Test

### **Null Hypothesis**

The means of the two groups are the same.

**Alternative Hypothesis** The means of the two groups are different.

**Demo**

One Way ANOVA

## Pairwise Tests

Compare each combination of groups

Suppose we reject the null hypothesis based on ANOVA and conclude that at least one mean group differs. How do we figure out which one? Post-hoc pairwise tests.

# Key Concepts

## Familywise Error Rate:

Probability that at least one test results in a false positive (Type I Error) conclusion.

## Bonferroni Correction:

Adjusts for multiple comparisons, reducing the chance of false-positive findings.

**Demo**

Pairwise Tests



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